

A PROJECT REPORT
ON
JOB PORTAL WEB APPLICATION

USING JAVA, SPRING BOOT AND ANGULAR

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INTRODUCTION

The Job Portal Web Application is a full-stack system designed to connect job seekers with employers. It provides a platform where job seekers can search for jobs and apply online, while employers can post job openings and manage applications.

This system simplifies the recruitment process by providing a centralized platform for both job seekers and recruiters. The application is built using Angular for the frontend and Spring Boot for the backend, ensuring a modern and scalable architecture

OBJECTIVE

Objectives include:

- To provide a centralized job searching platform
- To allow employers to post and manage jobs
- To enable job seekers to apply for jobs easily
- To implement secure authentication using JWT
- To improve user experience with a responsive interface

SCOPE

The scope of this project includes job posting, job searching, and application tracking. The system allows employers to manage job listings and view applicants, while job seekers can apply for jobs and track their applications.

The system can be extended in the future to include features like resume builder, job recommendations, and notifications

SYSTEM ARCHITECTURE

The system follows a 3-tier architecture:

1. Presentation Layer (Frontend - Angular)
2. Business Logic Layer (Backend - Spring Boot)
3. Data Layer (MySQL Database)

The frontend communicates with the backend using REST APIs, and the backend interacts with the database to store and retrieve data

TECHNOLOGIES USED

The following technologies are used in this project:

- Angular (Frontend Development)
- Spring Boot (Backend Development)
 - MySQL (Database)
- Spring Security (Authentication and Authorization)
 - JWT (Token-based Authentication)
 - HTML, CSS, TypeScript

MODULES

The system consists of the following modules:

1. Job Seeker Module:

- Register and login
- View jobs
- Search job
- By location
- Or name
- Apply for jobs

2. Employer Module:

- Post jobs
- Edit/Delete jobs
- View applicants

Database Design

The database consists of the following main tables:

- User Table (stores user details)
- Job Table (stores job information)
- Application Table (stores job applications)

Relationships:

- One employer can post multiple jobs
- One job can have multiple applications
- One user can apply to multiple jobs

IMPLEMENTATION

The implementation of the Job Portal Web Application is done using Angular for the frontend and Spring Boot for the backend. The system follows a REST API architecture where the frontend communicates with the backend to perform operations like job posting, job searching, and applying for jobs.

The backend handles business logic, authentication, and database operations, while the frontend provides an interactive user interface.

Below are some important code snippets used in the implementation:

The backend controller handles HTTP requests from the frontend

```
@RestController
```

```
@RequestMapping("/api/employer")
```

```
public class EmployerController {
```

```
    @PostMapping("/jobs")
```

```
    public String createJob() {
```

```
        return "Job Created Successfully";
```

```
    }
```

```
    @DeleteMapping("/jobs/{id}")
```

```
    public String deleteJob(@PathVariable Long id) {
```

```
        return "Job Deleted";
```

```
}  
}
```

Below is the actual service layer implementation used in the project

```
@Service
```

```
@RequiredArgsConstructor
```

```
public class JobService {
```

```
    private final JobRepository jobRepository;
```

```
    private final UserRepository userRepository;
```

```
    public void deleteJob(Long jobId, String employerEmail) {
```

```
        Job job = jobRepository.findById(jobId)
```

```
            .orElseThrow(() -> new RuntimeException("Job not found"));
```

```
        if (!job.getEmployer().getEmail().equals(employerEmail)) {
```

```
            throw new RuntimeException("You cannot delete this job");
```

```
        }
```

```
        jobRepository.delete(job);
```

```
    }
```

```
}
```

The frontend communicates with backend APIs using Angular services

```
applyToJob(jobId: number, formData: FormData) {  
    return this.http.post(  
        `${this.apiUrl}/jobseeker/apply/${jobId}`,  
        formData,  
        { responseType: 'text' }  
    );  
}
```

Angular components handle user interaction and UI logic

```
apply(jobId: number) {  
  
    const formData = new FormData();  
  
    formData.append("resume", new File(["Resume"],  
        "resume.pdf"));  
  
    this.jobService.applyToJob(jobId, formData).subscribe({  
        next: () => alert("Applied Successfully"),  
        error: () => alert("Error")  
    });  
}
```

CONCLUSION

The Job Portal Web Application successfully provides a platform for connecting job seekers and employers. It simplifies the recruitment process and improves efficiency.

The project demonstrates the use of modern technologies like Angular and Spring Boot to build a scalable and secure application

TESTING

The application was tested for various functionalities including login, job posting, job application, and data retrieval.

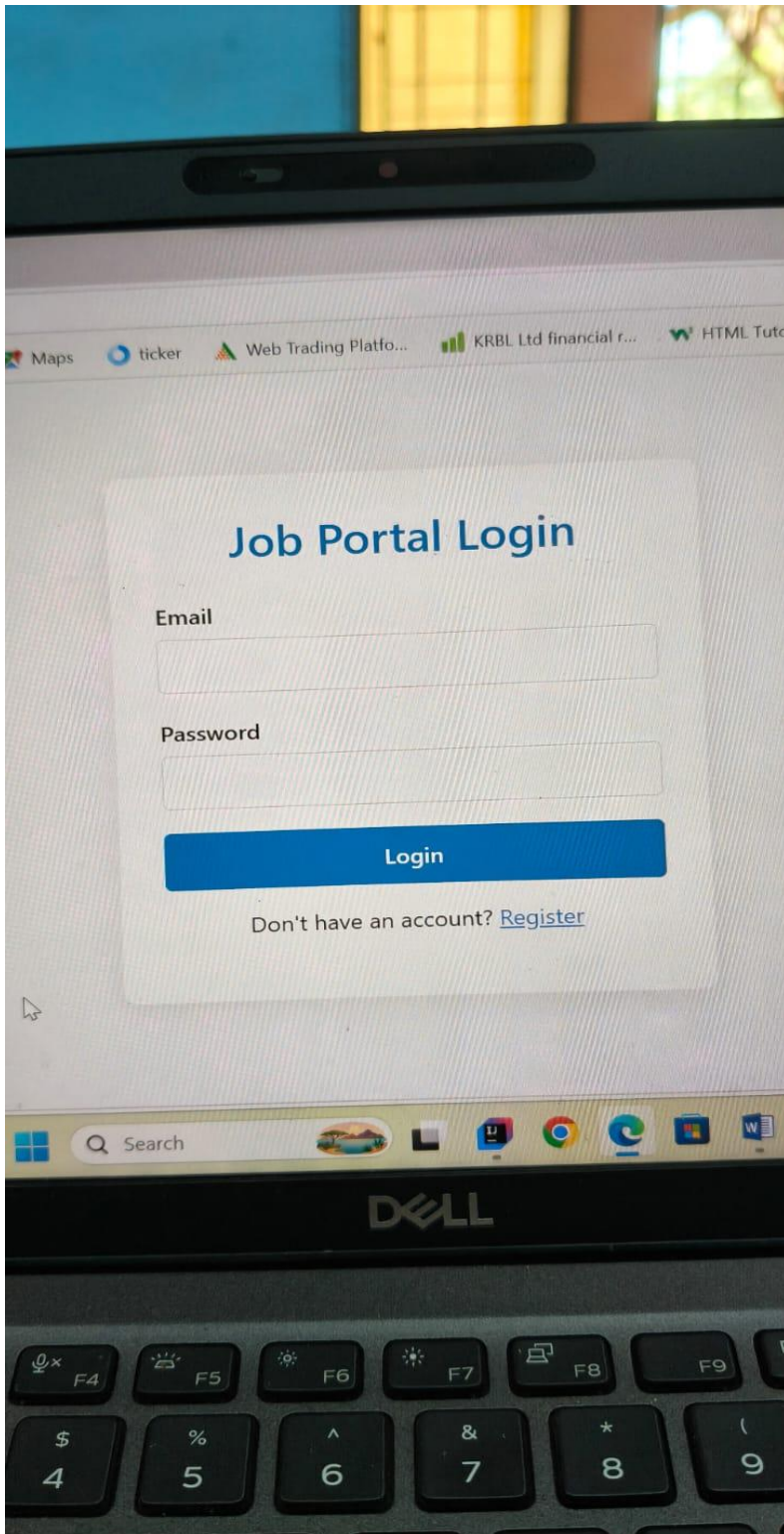
Different test cases were performed to ensure that the system works correctly and handles errors properly

FUTURE SCOPE

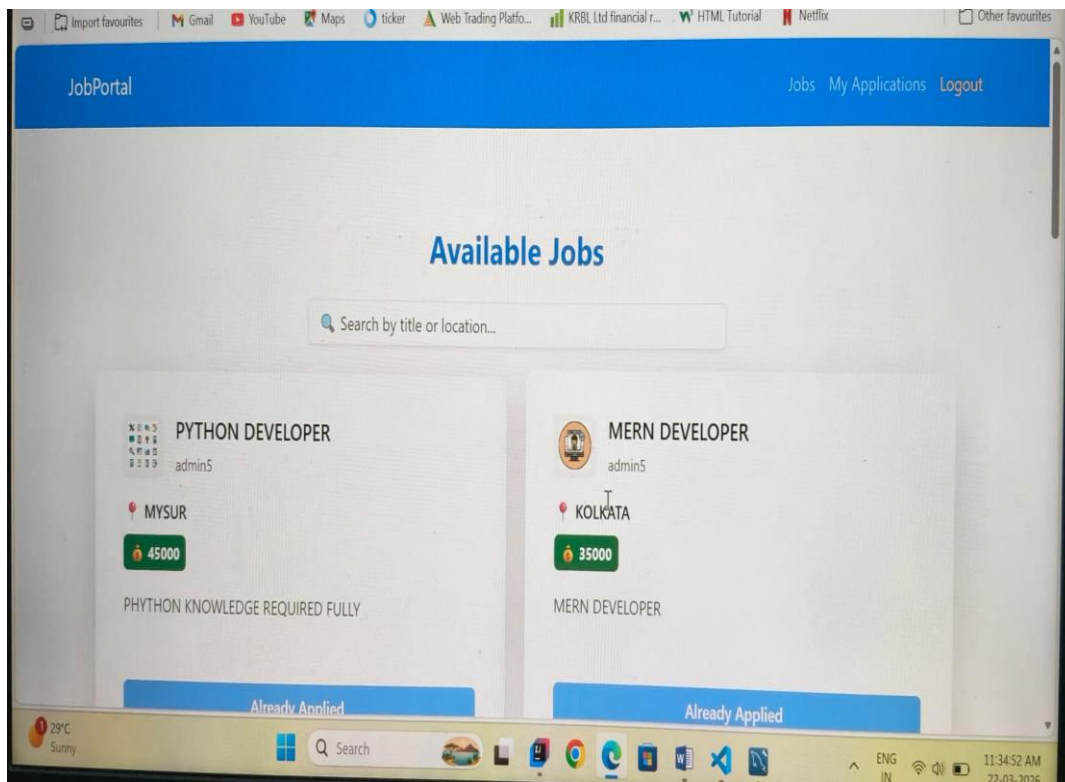
The project can be enhanced with additional features such as:

- Resume builder
- Email notifications
- Job recommendations using AI
- Mobile application version
- Advanced filtering options

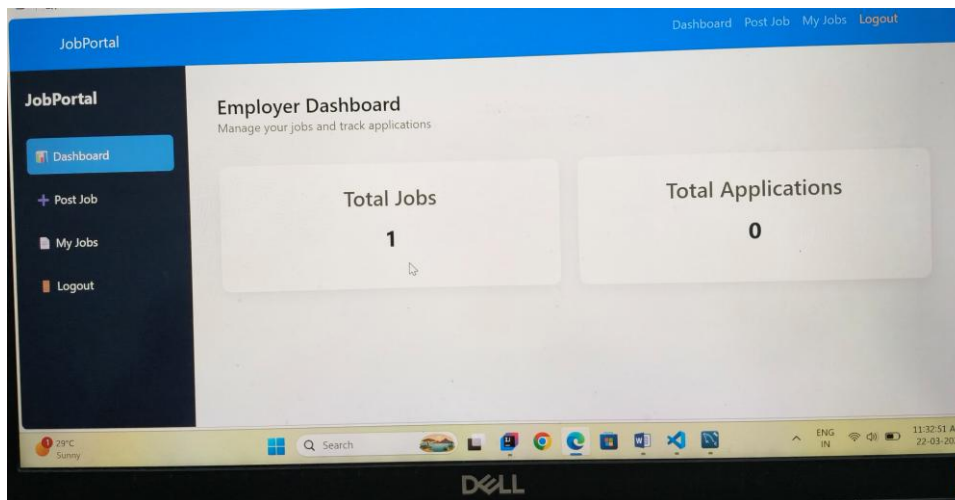
Login page



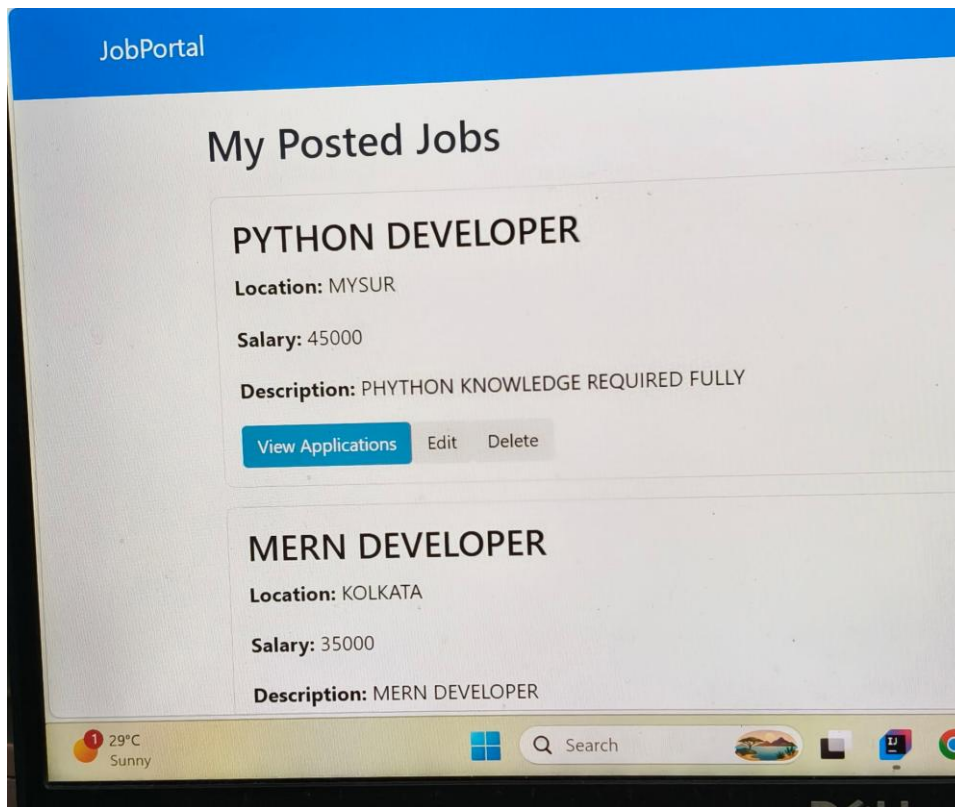
SEEKER SIDE



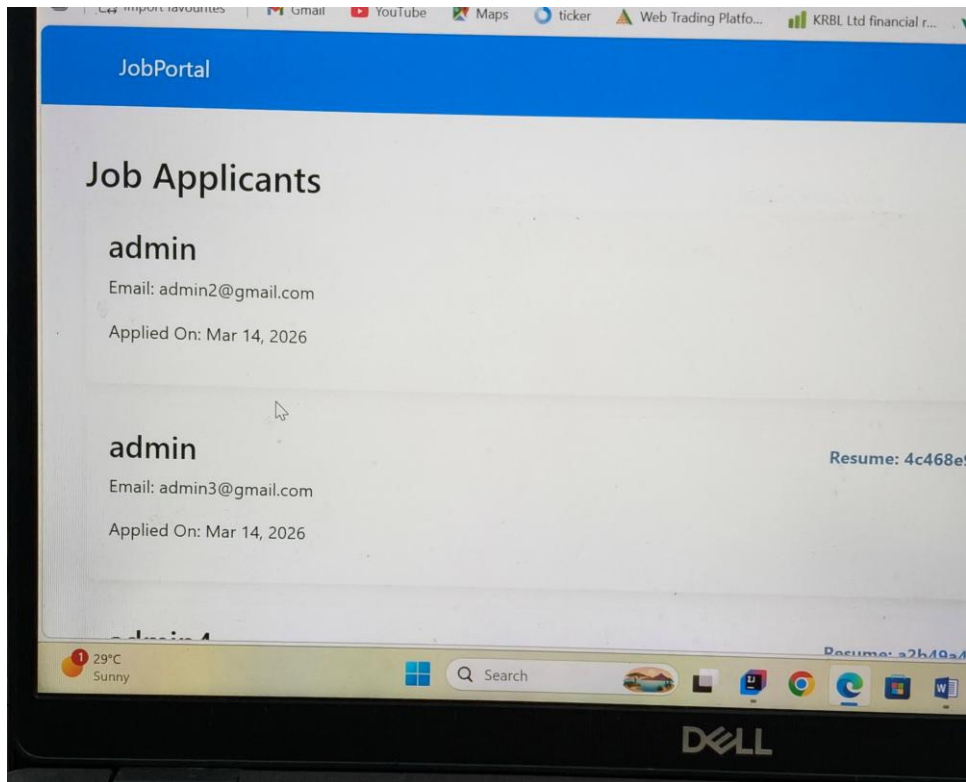
Employer Dashboard



VIEW POSTED JOB



VIEW APPLICANT



GITHUB & LINKEDIN

GitHub: <https://github.com/ansarihasan13-sudo>

LinkedIn: <https://www.linkedin.com/in/rashid-shaikh-425879237/>